

OncoEvidence: Mechanism-Guided Evidence Triage for Cancer Drug Repurposing

A completed-project report: when a knowledge graph earns its place, and how to show its mechanism reasoning is predictive and faithful

Akshatha Arunkumar

Final Project Deliverable | June 28, 2026

Summary. OncoEvidence ranks oncology drug-repurposing candidates on the PrimeKG knowledge graph, extracts the multi-hop mechanism path that would explain each one, and uses a literature-grounded language model to grade whether that mechanism is supported. The project is built around an honest question: once a strong tabular ranker is in place, what does the graph actually add? The answer is mechanism, and this report goes further than earlier drafts by showing that the graph’s mechanism signal is (i) *predictive* over time and (ii) *faithful* to curated biology, with one orthogonal validation reported as an honest negative.

Six findings.

1. On plain link ranking, a tuned XGBoost matches or beats the graph network in every regime, so ranking alone does not justify the graph.
2. Only the graph produces a traceable multi-hop mechanism path.
3. Mechanism structure separates true oncology indications from random pairs at AUROC 0.879, holds up under hard negatives, and agrees with curated DrugMechDB at 0.802.
4. Blinded mechanism recovery: with the direct drug-target edge hidden, only the jointly trained GNN names the curated bridge gene (Recall@10 0.25), while trivial and link-only baselines collapse to about zero.
5. **New: prospective validation.** On a time-split test, the GNN ranks *future* oncology indications at AUROC 0.930 and beats a structure-blind control by +0.047 AUROC.
6. **New: faithfulness.** Deleting a pair’s curated mechanism edge drops the bridge gene’s score far more than deleting a random edge (contrast 0.124, $p \approx 3.6 \times 10^{-34}$, faithful in 83% of cases).

1 Background and significance

Developing a new drug takes many years and large cost, while thousands of approved drugs already have known safety profiles. If an approved drug acts on a protein or pathway that also drives a cancer, repurposing it can be a fast route to a new treatment. Biomedical knowledge graphs such as PrimeKG encode millions of relationships among drugs, proteins, genes, pathways, and diseases, so the information needed to reason about such mechanisms exists as paths through the graph. The open problem is *trust*: a bare drug-disease score hides its reasoning and cannot separate a real mechanism from a coincidence.

Novelty, stated carefully. Knowledge-graph repurposing with explanations is not new: TxGNN is a graph foundation model with multi-hop rationales, and KGML-xDTD predicts treatment links with KG-based mechanism paths. This project does not claim “a knowledge graph plus a language model for repurposing” as novel. The defensible contribution is methodological:

quantify *when* graph structure adds value beyond a strong tabular ranker, show that its value is mechanism extraction rather than better ranking, and then audit those mechanisms three ways that most demos skip: against curated mechanism databases, against a prospective time-split, and against a counterfactual faithfulness test.

2 Datasets and system

Resource	Content	Role
PrimeKG	~129k nodes, ~8.1M directed edges; 7,957 drugs, 17,080 diseases (~3,170 oncology)	Ranking + mechanism paths
Europe PMC	abstracts and open-access full text via REST	Evidence + time axis
DrugMechDB	curated drug mechanism-of-action paths (UniProt-keyed)	Mechanism ground truth
ClinicalTrials.gov	registered interventional studies (API v2)	Orthogonal check

Table 1: Public resources. DrugMechDB UniProt accessions are mapped to HGNC symbols via mygene.info; the per-pair time axis in Finding 5 is the earliest Europe PMC co-mention year.

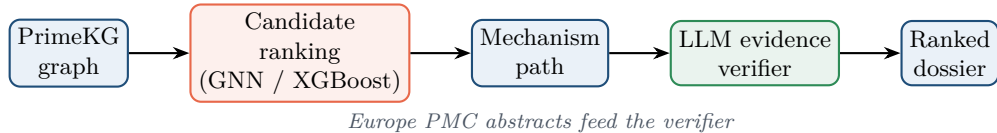


Figure 1: The OncoEvidence pipeline: rank candidates, extract a mechanism path, verify it against the literature, output an evidence-backed shortlist.

3 Methods

Ranking. Repurposing is link prediction on drug-indication-disease edges. A heterogeneous GraphSAGE encoder builds node embeddings by message passing,

$$h_v^{(l)} = \sigma\left(W^{(l)}\left[h_v^{(l-1)} \parallel \text{AGG}_{u \in \mathcal{N}(v)} h_u^{(l-1)}\right]\right), \quad s(d, c) = \text{MLP}([z_d \| z_c]). \quad (1)$$

The GNN, a DistMult embedding, and a tuned XGBoost all receive the *same* SentenceTransformer node features, so any gap measures graph structure, not features.

Mechanism extraction. For a candidate we search PrimeKG for short paths through mechanism relations, scoring each by specificity and penalizing promiscuous hubs:

$$\text{score}(p) = \beta_{\text{type}} + \frac{1}{\log_2(\deg(\text{bridge}) + 2)} \cdot \frac{1}{\log_2(\delta_{\text{drug}} + 2)}, \quad (2)$$

where δ_{drug} is the number of drugs that bind the bridge protein (an inverse-frequency penalty that suppresses carriers such as albumin) and β_{type} orders direct-target over interaction over shared-pathway.

Evidence verification. For the top path we retrieve abstracts, extract sentences mentioning both drug and gene, and ask a language model to grade the mechanism supported / weak / contradicted / unknown, requiring a verbatim quote and grading a purely statistical association as weak.

Mechanism head. A joint objective adds a contrastive term that ranks the curated Drug-MechDB bridge gene above degree-matched decoys, using the same encoder embeddings, $s_{\text{mech}}(d, g, c) = \text{MLP}([z_d \| z_g \| z_c])$. This is the head used in Findings 4 and 6.

4 Results

4.1 Finding 1: the link task does not need the graph

With a tuned XGBoost on shared features, the GNN does not win in any regime (Table 2). For link ranking with rich text features, a strong tabular model suffices; a drug/disease name embedding lets it take a semantic-similarity shortcut without graph reasoning.

Test AUROC (5 seeds)	GNN	XGBoost	MLP	KGE
Transductive	0.977	0.988	0.973	0.907
Inductive (cold-disease, oncology)	0.882	0.963	0.930	0.452
Inductive (cold-drug)	0.956	0.958	0.931	0.482

Table 2: Corrected benchmark. The DistMult memorizer (KGE) collapses on unseen nodes, confirming the task needs content features, not memorized identity.

4.2 Finding 2: the graph carries real mechanism

A tabular model cannot produce a traceable mechanism; the graph can. On true oncology indications the extractor recovers textbook direct-target mechanisms, for example Quizartinib-FLT3 (myeloid leukemia), Tamibarotene-RARA (acute promyelocytic leukemia), Topotecan-TOP1 (ovarian cancer), and Trifluridine-TYMS (colorectal cancer).

4.3 Finding 3: mechanism structure separates true from random pairs

Over 400 true indications versus 400 random pairs, the mechanism signal separates the two groups at **AUROC 0.879** (Table 3). The extracted bridge genes also agree with curated DrugMechDB mechanisms for 211 of 263 covered pairs (**agreement 0.802**). The language-model verifier is the conservative reviewer: of its “supported” calls that DrugMechDB covers, 6/7 are curated-confirmed, versus 13/22 for a keyword baseline (small denominators, reported as a pilot signal).

Metric	True	Random
Mean mechanism score	1.95	0.17
Direct-target rate	34.0%	0.25%
Any mechanistic path	81.3%	8.8%

Separation AUROC
0.879

Table 3: Mechanism paths separate true indications from random pairs. A true indication is roughly 136× more likely to have a direct drug-target to cancer-gene link.

4.4 Finding 4: the graph’s one genuine ranking-free edge

We trained the GNN jointly and asked it to name the curated bridge gene for held-out cold-disease pairs (3 seeds, $n \approx 186\text{--}260$). Unblinded, a trivial “the drug’s own targets” baseline dominates

because the bridge gene is usually the drug’s direct target. But with that direct edge hidden (mechanism-blinded), the trivial and degree baselines collapse to about zero while the joint GNN still recovers the gene from indirect structure (Table 4). A link-only GNN scored through the identical head recovers nothing, so the signal is attributable to the auxiliary objective.

Bridge-gene recovery	Unblinded R@10	Blinded R@10	Blinded MRR
Joint GNN (link + mechanism)	0.31	0.25	0.21
Trivial target-lookup	0.80	0.00	0.00
Link-only GNN (same head, no aux loss)	0.00	0.00	0.00
Degree prior	0.02	0.02	0.02

Table 4: Mechanism recovery on held-out pairs. When the direct edge is hidden, only the joint GNN recovers the gene, a small (~ 0.25 R@10) but genuine graph-only signal. Caveat: this is recovery of *known* mechanism via indirect paths, not de novo discovery, on a modest sample.

4.5 Finding 5 (new): the model is predictive over time

The findings above are retrospective. To test whether the model could have flagged an indication *before* it was established, we assign each true oncology pair an approximate first-evidence year (earliest Europe PMC co-mention), choose cutoff $T = 2005$, keep only pairs known by T inside the message-passing graph, and hold out the later pairs as a prospective test set (350 sampled pairs; 341 years resolved; 240 past, 101 future; 3 seeds). The GNN ranks the held-out future indications at **AUROC 0.930** and **AUPRC 0.707**, and beats the structure-blind control on every metric (Table 5). Unlike the plain link task in Finding 1, the graph adds a small but consistent edge here, widest on the harder metrics (AUPRC +0.115, recall@100 +0.116).

Prospective ($T=2005$)	AUROC	AUPRC	R@50	R@100
GNN (graph)	0.930	0.707	0.393	0.703
MLP (structure-blind)	0.882	0.592	0.337	0.587
Graph gain	+0.047	+0.115	+0.056	+0.116

Table 5: Prospective time-split (mean over 3 seeds). The model ranks future indications well above chance, and the graph beats the tabular-style control on every metric and seed. Caveat: the year proxy is the earliest literature co-mention, not a regulatory date, and this is a single seeded 350-pair sample, so magnitudes are indicative.

4.6 Finding 6 (new): the mechanism reasoning is faithful

Can we trust the mechanism the graph names? We take the trained joint model and, for each held-out pair whose bridge gene is a real drug-target edge, recompute the gene’s mechanism score with the full graph, with that pair’s curated mechanism edge deleted, and with a random other target edge of the same drug deleted (316 instances, 2 seeds, no retraining). Removing the curated edge degrades the bridge gene far more than removing a random one (Table 6): a paired Wilcoxon test gives $p \approx 3.6 \times 10^{-34}$, the model is faithful in 83% of instances, and the two populations separate at AUROC 0.835. This turns “the graph explains” from a claim into a measured property.

Counterfactual edge removal	Score drop	Rank drop
Remove curated mechanism edge	0.124	~211
Remove random target edge (same drug)	-0.001	~2
Contrast (faithfulness)	0.124	—

Table 6: Faithfulness by counterfactual edge removal (pooled, 316 instances). Faithful in 83.3% of cases; paired Wilcoxon $p \approx 3.6 \times 10^{-34}$; separation AUROC 0.835. Measured on the well-posed population (pairs whose bridge gene is an actual drug-target edge), in logit space.

4.7 An honest negative: orthogonal trial validation

We also asked whether top *novel* predictions (not already PrimeKG indications) appear as registered oncology trials more than matched controls. Top raw-score predictions were not enriched over random (6.7% versus 7.4%, Fisher $p = 0.69$). Raw score does track trial existence above chance (AUROC 0.676 over 810 pairs, $p \approx 1.5 \times 10^{-5}$), but this is a popularity confound: one broadly-indicated drug (folic acid) accounts for 78% of the top set’s hits. The specificity-lift ranking that de-confounds popularity shows no enrichment at all ($0.55\times$ versus random, $p = 0.97$). We report this as inconclusive: the raw scores track human trial activity mostly through popular drugs, not through the specific novel shortlist. A usable version needs MeSH/ChEBI name normalization and trial phase/recency, which is future work.

5 Discussion

The contribution is a disciplined account of where a knowledge graph helps. On plain link ranking it does not beat a tuned tabular model, and we say so. Its value is the traceable mechanism a tabular model cannot produce, and this report strengthens that case along the two axes that matter for a real triage tool. Finding 5 shows the mechanism-aware graph model is *predictive*: trained only on pre-2005 structure, it ranks post-2005 indications at 0.930 AUROC and beats the structure-blind control, so the signal is not merely retrospective bookkeeping. Finding 6 shows the mechanism reasoning is *faithful*: deleting the curated edge, and only that edge, collapses the prediction, so the explanation reflects the biology the model actually used.

Limitations. The blinded recovery and faithfulness tests are recovery of known mechanism, not de novo discovery; some samples are modest; the time axis is a literature proxy; and the orthogonal trial check was a negative. None of the predictions are medical advice.

6 Reproducibility

Open tools (PyTorch Geometric, scikit-learn, XGBoost) and public data (PrimeKG, Europe PMC, DrugMechDB, ClinicalTrials.gov). Every reported number has a script and a JSON record in `results/`: `oncorepurpose.json` (Finding 1), `mechanism_eval.json` (Finding 3), `mechanism_recovery_eval.json` (Finding 4), `temporal_split_eval.json` (Finding 5), `edge_faithfulness_eval.json` (Finding 6), and `clinicaltrials_validation.json` (negative). A six-part Colab notebook track reproduces the pipeline end to end, with Part 4 fully self-contained. All predictions are hypothesis-generating research, not medical advice.

References

- [1] Huang K, Chandak P, et al. *A foundation model for clinician-centered drug repurposing (TxGNN)*. Nature Medicine, 2024.
- [2] Chandak P, Huang K, Zitnik M. *Building a knowledge graph to enable precision medicine (PrimeKG)*. Scientific Data, 2023.
- [3] *KGML-xDTD: a knowledge graph-based framework for drug treatment prediction and mechanism description*. 2023.
- [4] Zitnik M, Agrawal M, Leskovec J. *Modeling polypharmacy side effects with graph convolutional networks (Decagon)*. Bioinformatics, 2018.
- [5] Mayers M, et al. *DrugMechDB: a curated database of drug mechanism paths*. 2022.
- [6] Hamilton W, Ying R, Leskovec J. *Inductive representation learning on large graphs (GraphSAGE)*. NeurIPS, 2017.